

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 2.3 TRIGONOMETRY: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for our unit. Driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment, without ever measuring it directly? Answer every section in the space under each prompt. Use what you observed and calculated in our lessons: the school water tank, Amina and Baraka, your paper triangles, your own clinometer and your outdoor measurements. Show your working for every calculation, include units, and write full sentences when you are asked to explain. You may use a calculator. Where a question gives a value from the calculator, you do not need to find it again.

Part 1: The triangle hidden in the water-tank problem

Amina stands 10 m from the base of the school water tank and sights the top of the tank. Draw and label a clear sketch of this situation. Mark the angle of elevation, and label the three sides of the right-angled triangle using the words hypotenuse, opposite and adjacent. Then say which of the three sides you can measure directly with a tape measure from the ground, and which side you cannot measure directly. Explain why 'opposite' and 'adjacent' would swap names if you chose the other acute angle as your reference angle, but 'hypotenuse' would not change.

Sketch: a vertical line for the tank tower (height h), a horizontal ground line of 10 m from Amina to the base of the tower, and a sloping line of sight from Amina's eye or position to the top of the tank. The right angle is at the base of the tower. The angle of elevation (38°) is marked at Amina's position. Labels relative to the angle of elevation: the tank height is the OPPOSITE side, the 10 m ground distance is the ADJACENT side, and the line of sight is the HYPOTENUSE. You can measure the ground distance (adjacent) with a tape measure; you cannot measure the tank height (opposite) directly because you cannot climb the tank. Opposite and adjacent depend on which acute angle you are looking from: the side that faces one acute angle is the side that lies next to the other acute angle, so when the reference angle moves, the two names swap. The hypotenuse is always the side opposite the right angle and the longest side, so it keeps its name whichever acute angle you choose.

Part 2: Calculating the height from two positions

In Lesson 1, Amina stood 10 m

(a) $\tan \theta = \text{opposite} \div \text{adjacent} = \text{height} \div \text{horizontal distance}$, so $h = d \times \tan \theta$. (b) Amina: $h = 10 \times \tan 38^\circ = 10 \times 0.781 = 7.81 \text{ m}$,

from the base of the tank and measured an angle of elevation of 38°. Baraka stood 20 m from the base and measured an angle of elevation of 22°. Your calculator gives $\tan 38^\circ \approx 0.781$ and $\tan 22^\circ \approx 0.404$. (a) Write down the tangent ratio in words for the angle of elevation in this problem and rearrange it to give a formula for the height h. (b) Use your formula to calculate the height of the tank from Amina's measurements and from Baraka's measurements. Give each answer to 1 decimal place. (c) Compare your two heights. Would you expect them to be exactly equal? Explain your thinking, and say what could cause any difference you find.	which is 7.8 m to 1 decimal place. Baraka: $h = 20 \times \tan 22^\circ = 20 \times 0.404 = 8.08$ m, which is 8.1 m to 1 decimal place. (c) The two heights are very close (about 0.3 m apart). Both students are finding the height of the same tank, so we would expect the same answer, but we should not expect them to be exactly equal. The angles were measured by hand with a simple device, so each angle has a small measurement uncertainty, and the calculator values were rounded. The difference comes from measurement uncertainty and not from an error in the mathematics. Both answers show that the tank is about 8 m tall, even though the two students stood at different distances and recorded different angles.
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Part 3: Why does the method work from any distance?	
In Lesson 4 you drew three right-angled triangles that each had a 35° angle but had adjacent sides of 5 cm, 8 cm and 11 cm. You found that $\text{opp} \div \text{hyp}$, $\text{adj} \div \text{hyp}$ and $\text{opp} \div \text{adj}$ stayed almost the same in all three triangles, even though the triangles were different sizes. The calculator gave $\sin 35^\circ \approx 0.574$, $\cos 35^\circ \approx 0.819$ and $\tan 35^\circ \approx 0.700$. (a) Match each of the three ratios to its name (sine, cosine, tangent) and write the mnemonic that helps you remember them. (b) Use this finding to explain why the tangent of an angle can be used to find the tank height from any distance. In your answer, say what stays the same for every	(a) $\text{opp} \div \text{hyp}$ is sine (sin), $\text{adj} \div \text{hyp}$ is cosine (cos), and $\text{opp} \div \text{adj}$ is tangent (tan). The mnemonic is SOH-CAH-TOA: Sine = Opposite/Hypotenuse, Cosine = Adjacent/Hypotenuse, Tangent = Opposite/Adjacent. (b) For a fixed angle, the ratio opposite $\div$ adjacent is always the same number, whatever the size of the triangle, because triangles with the same angle are bigger or smaller copies of each other. So a given angle always has one fixed tangent value, and that value links the height to the ground distance: $\text{height} = \tan \theta \times \text{ground distance}$. Students at different distances measure different angles, so their tan values are different (for example $\tan 38^\circ \approx 0.781$ and $\tan 22^\circ \approx 0.404$). What stays the same for every student is the rule $h = d \times \tan \theta$ and the height of the tank itself. A student who is further away has a bigger distance but a smaller angle and so a smaller $\tan \theta$, and the product $d \times \tan \theta$ gives the same height of the tank. Because the ratio depends only on the angle and not on the size of the triangle, the method works from any distance. (c) Our drawn triangles were measured with a ruler and protractor, so small measurement errors in the lengths and the angle make our ratios slightly different. The calculator holds the true ratio, so the differences are measurement error and not a failure of the mathematics.

student and what is different for each student. (c) Your measured ratios were slightly different from the calculator values. Give one reason why.	
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Part 4: The complementary-angle shortcut

The two acute angles in a right-angled triangle add up to 90°. In the water-tank triangle, Amina's angle of elevation is 38°. (a) What is the size of the other acute angle, at the top of the triangle? (b) Using the words opposite and adjacent, explain how the tank height is related to each of these two angles. (c) Use your answer to (b) to explain why $\sin 38^\circ$ and $\cos 52^\circ$ must have the same value. (d) In the table you completed in Lesson 6 you found the difference $\sin \theta - \cos(90^\circ - \theta)$ for each pair of angles. What value did you find in every row, and what does this show?	(a) $90^\circ - 38^\circ = 52^\circ$. (b) The tank height is OPPOSITE the 38° angle and ADJACENT to the 52° angle. The line of sight is the hypotenuse for both angles. (c) $\sin 38^\circ = \text{opposite} \div \text{hypotenuse} = \text{tank height} \div \text{line of sight}$. $\cos 52^\circ = \text{adjacent} \div \text{hypotenuse}$, and the side adjacent to 52° is the same tank height, so $\cos 52^\circ = \text{tank height} \div \text{line of sight}$ as well. Both ratios use the same pair of sides, so $\sin 38^\circ = \cos 52^\circ$ (both about 0.616). This is the general result $\sin \theta = \cos(90^\circ - \theta)$, and also $\cos \theta = \sin(90^\circ - \theta)$. (d) The difference was 0 in every row. This shows that the sine of an angle equals the cosine of its complementary angle for every pair, so knowing one ratio immediately gives the other without measuring or calculating again.
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Part 5: Closing the loop: our outdoor measurement and why trigonometry matters

In Lesson 8 you used your own clinometer at two stations, Observer A at 8 m and Observer B at 12 m from the base of the school water-tank tower. (a) Describe the steps you followed to find the height of the tank without climbing it. (b) At which station was your angle of elevation larger, and why? (c) Write a short explanation in CER form (Claim, Evidence, Reasoning) that answers the driving question: how can we	(a) 1. Stand at a marked station and measure the horizontal distance from the base of the tower (8 m for Observer A, 12 m for Observer B). 2. Sight the top of the tank through the straw of the clinometer and read the angle of elevation from the string and plumb bob on the protractor. 3. Use $h = d \times \tan \theta$ with that station's distance and angle to calculate the height. 4. Repeat at the second station and compare the two heights. (b) The angle was larger at the 8 m station (Observer A), because the closer you stand, the steeper you must look up to see the top of the tank. At 12 m the line of sight is less steep, so the angle is smaller. (c) Claim: the height of the tank can be found by measuring the horizontal distance and the angle of elevation, then using $h = d \times \tan \theta$. Evidence: at each station I recorded the distance (8 m and 12 m) and the angle from my clinometer, and my two calculated heights were nearly the same, with only a small difference. Reasoning: $\tan \theta = \text{opposite} \div \text{adjacent} = \text{height} \div \text{distance}$, and $\tan \theta$ is a constant ratio for a fixed angle, so multiplying the distance by $\tan \theta$ gives the same height from any position. The small difference comes from measurement uncertainty in the angles, not from an error in the mathematics. (d) Trigonometry is important because it lets us find heights and distances that we cannot measure directly. Kenyan surveyors use it to map the Rift Valley highlands, and foresters in the Mau Forest use it to measure the height of trees without felling or climbing them. KeNHA engineers also use it in designing the Thika Superhighway overpasses.
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find the tank height from a measured distance and an angle? Use your own results from the two stations as your evidence. (d) The KICD key inquiry question asks: Why is trigonometry important? Give your answer using two examples of people who use this method in Kenya.	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Naming the sides and using SOH-CAH-TOA	Labels opposite, adjacent and hypotenuse correctly from the chosen angle, explains why the names swap, and states sine, cosine and tangent with the SOH-CAH-TOA definitions accurately.	Labels the sides correctly for the water-tank triangle and states the three ratios, but the explanation of the swapping of names is incomplete.	Mislabeled sides or confuses the ratios, and gives little or no explanation of the reference angle.
Calculation with $h = d \times \tan \theta$	Rearranges the tangent ratio correctly, calculates both heights accurately (about 7.8 m and 8.1 m) with units, and shows all working.	Uses the formula correctly with only a minor slip in rounding, units or working.	Does not rearrange the formula correctly or makes calculation errors that change the result.
Explaining consistency and the constant ratio	Clearly explains that the ratio depends only on the angle, that different distances give different angles and tan values but the same height through $h = d \times \tan \theta$, and attributes small differences to measurement uncertainty.	States that the ratio is constant and the heights are similar, but the reasoning about why is partly stated.	Says only that the answers are 'the same' or 'different' with no mathematical reasoning.
Complementary angles and CER explanation with Kenyan contexts	Explains $\sin \theta = \cos(90^\circ - \theta)$ using opposite and adjacent, and writes a full CER using the 8 m and 12 m results, with two accurate Kenyan examples of use.	Gives the complementary-angle result and a CER with claim and evidence, but the reasoning or the examples are thin.	States the result without explanation, and the CER is missing parts or does not use the outdoor evidence.

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